

A P R O P O S A L F O R W O R L D L I B E R T Y F I N A N C I A L

AGENTIC FINANCE

USD1 × AUTONOMOUS AGENTS

AGENTIC PAYMENTS BOT · x402 · MPP · AP2 · A2A · MCP · ERC-8004

SENTIENT-AGI · github.com/sentient-agi/agnostic-payments-bot

▷ 2026

THE THESIS

WHY THIS · WHY NOW · WHY WLFI

\$4.5B

USD1 in circulation

#4 stablecoin · fastest-growing of the period

\$600M

x402 annualized volume

154M machine payments across Base + Solana

97M

MCP SDK pulls / month

every major agent client speaks the stack

AGENTS ARE THE NEXT CUSTOMERS

AI agents are becoming the primary interface for every business — WLFI's own AgentPay vision. They transact 24/7, at machine speed, in machine sizes.

THE MACHINE ECONOMY NEEDS A COMPLIANT DOLLAR

x402 and MPP settle in stablecoins. A regulated, treasury-backed, attestation-audited dollar is the natural reserve asset — that dollar is USD1.

WE BRING THE MISSING LAYER

A production agent payment skill: three protocols (x402 · MPP · AP2) as both client and server, many payment backends, policy enforcement, on-chain trust — Apache-2.0.

PART I

WORLD LIBERTY FINANCIAL

▷ WHERE DEFI MEETS TRADFI

WORLD LIBERTY FINANCIAL

COMPANY SNAPSHOT — OUR READ OF YOUR PLATFORM

***"Where DeFi meets TradFi"** — bridging legacy finance and the open economy with purpose-built, onchain products, for institutions and individuals alike.*

REGULATED CORE

USD1 issued via BitGo Trust Company (regulated SD trust). Reserves: cash + short-term US Treasuries in government money-market funds. Monthly attestations + real-time proof of reserves.

COMMUNITY GOVERNED

WLFI token holders propose, review, and vote — the token became tradable through a governance vote. Institutional posture with an open-governance base.

MULTICHAIN BY DESIGN

Ethereum · BNB Chain · Solana, expanding via Chainlink CCIP. Bridge and convert tools ship as first-class products, not afterthoughts.

AGENT-FORWARD

AgentPay SDK already live: policy enforcement, human approval, plugin architecture, TEE / Secure Enclave signing. The strategic intent is declared.

Miami, FL · fintech + DeFi protocol · flagship: USD1 digital dollar

PRODUCT PORTFOLIO

SIX SURFACES — ONE ONCHAIN FINANCE STACK

◆ USD1

US-dollar stablecoin for a new era of finance
— stable, secure, transparent by design.
Zero-fee mint & redeem.

▲ WLFI TOKEN

Governance asset — now tradable by
community vote. Steers the platform across
ETH · BSC · Solana.

■ WLFI MARKETS

Supply assets for rewards or borrow against
collateral — lending infrastructure provided
by Dolomite.

● AGENTPAY SDK

Open SDK: agents hold funds and pay with
policy enforcement and built-in human
approval.

◎ WLFI APP

Coming soon — use crypto via wallet or
bank account; spend by accessing liquidity
anytime.

★ BRIDGE & CONVERT

Move USD1 / WLFI across integrated
networks; swap in and out of the ecosystem.

The assets, the rails, and the retail surface exist — the agent layer is the next chapter.

USD1 — THE ASSET

A REGULATED DIGITAL DOLLAR AT SCALE

\$4.5B

circulating supply

crossed in Q1 2026

#4

stablecoin by mkt cap

surpassed PayPal's PYUSD

\$2B

MGX → Binance deal

settled officially in USD1

TRUST STRUCTURE

Issued and legally managed by BitGo Trust Company; reserves in cash + short-duration Treasuries; dynamic liquidity managed with Fidelity money-market funds.

COMPLIANCE POSTURE

Designed for GENIUS-Act-style frameworks: monthly reserve disclosures, audits, AML/KYC via regulated custody — the profile institutions can sign off on.

DISTRIBUTION

Native on Ethereum, BNB Chain, Solana with Chainlink CCIP interoperability; deep exchange integrations and a growing lending footprint.

THE STRATEGIC GAP

EVERY PRODUCT SERVES HUMANS – THE NEXT CUSTOMERS ARE MACHINES

WHAT EXISTS

- ▶ USD1 — a compliant settlement asset at \$4.5B scale
- ▶ Markets, bridge, convert — the liquidity rails
- ▶ WLFI App — the human retail surface (soon)

THE SIGNAL

- ▶ AgentPay SDK ships: policy engine, approvals, TEE signing
- ▶ “AI agents are becoming the primary interface for every business”
- ▶ Open-by-default stance — plugins, adapters, integrations

THE GAP

- ▶ No agent stack speaking x402 / MPP / AP2 — as client or server
- ▶ No A2A discovery or ERC-8004 on-chain trust for agents
- ▶ No MCP context plane feeding rates & data into policy

The proposal: close the gap with one open-source skill – and make USD1 the default settlement asset of the agent economy.

PART II

THE PROPOSAL

▷ AGENTIC PAYMENTS BOT · SENTIENT-AGI

AGENTIC PAYMENTS BOT

TRIPLE PROTOCOL · CLIENT & SERVER · OPEN AGENT SKILLS ECOSYSTEM

A triple-protocol (x402 + MPP + AP2) agentic payment service. Agents autonomously initiate, validate, and execute payments across web3 and web2 rails — and monetize their own services on the same protocols. AWS KMS key custody, a deterministic policy engine, an immutable audit trail, human-in-the-loop confirmation. Apache-2.0 · TypeScript · Docker-ready · ships as a SKILL.md for any agent client.



TRIPLE PROTOCOL

x402 + MPP + AP2 behind one paymentIntent schema — best rail wins



CLIENT + SERVER

Pays for services and charges for its own — both sides of every rail



MULTI-BACKEND

Viem for EVM chains · Solana · Stripe · PayPal · Visa · Mastercard — one router



POLICY ENGINE

Per-tx limits · daily / weekly / monthly aggregates · blacklists · time windows



KMS + AUDIT

AWS KMS ECDSA signing — keys never leave the HSM; SQLite immutable log



HUMAN-IN-THE-LOOP

Configurable confirmation thresholds — agent alone or with approval

Runs in: Claude Code · Codex · Junie · OpenCode · GitHub Copilot · Gemini CLI · OpenClaw — one skill, every client

THREE PROTOCOLS × TWO ROLES

EVERY RAIL, BOTH DIRECTIONS — THE BOT PAYS AND GETS PAID

x402

AS CLIENT ▼

Detects 402 challenges, signs payment payloads via KMS, retries with PAYMENT-SIGNATURE — consumes any paid API.

AS SERVER ▲

Gates the agent's own endpoints behind HTTP 402 — monetizes skills per call, settles in stablecoins.

MPP

AS CLIENT ▼

Opens sessions, streams signed micropayments against a pre-authorized budget — thousands of calls, one settlement.

AS SERVER ▲

Issues sessions and meters usage — sells inference, data, and agent work by the unit.

AP2

AS CLIENT ▼

Builds and signs Intent Mandates; executes charges through web2 gateways under user delegation.

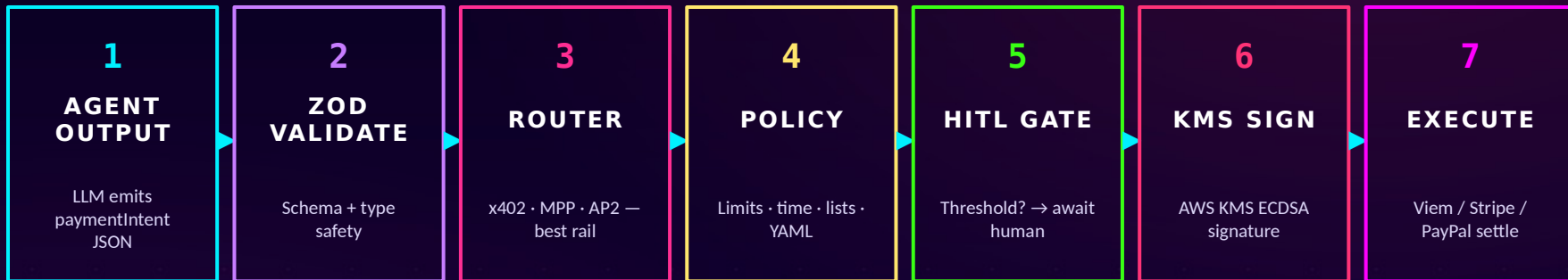
AS SERVER ▲

Accepts mandates as a merchant — a verifiable evidence chain behind every sale.

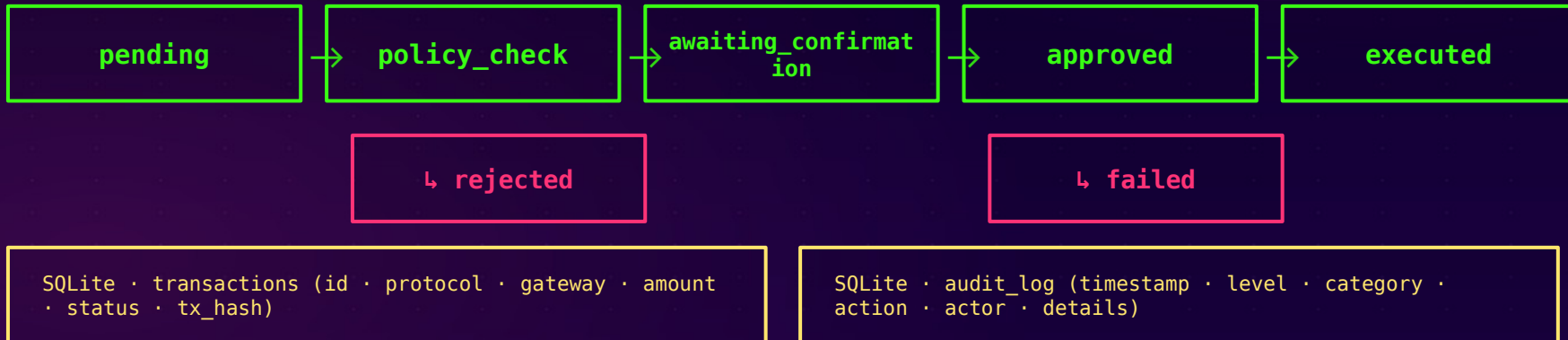
PAYMENT BACKENDS — Viem (Ethereum · BNB Chain · Base · Polygon · Arbitrum) · Solana · Stripe · PayPal · Visa · Mastercard — one router, one policy engine, one audit trail.

THE PIPELINE

FROM AGENT INTENT TO SETTLED PAYMENT — EVERY STEP GATED



TRANSACTION STATE MACHINE



POLICY IS COMPLIANCE

THE ENGINE SPEAKS TRADFI NATIVELY — A CONTROL FOR EVERY CONTROL

Per-tx · daily · weekly · monthly limits



Exposure & velocity limits — position caps an examiner recognizes

Blacklists · destination allowlists



Sanctions & counterparty screening (OFAC-style lists as YAML)

Time-window rules



Trading-hours controls & change-freeze windows

HITL confirmation thresholds



Four-eyes / maker-checker approval above materiality

Immutable SQLite audit_log



Examinable audit trail — every decision, actor, and transition

NATIVE FIT WITH AGENTPAY — WLF1's own policy TUI reads: USD1 per-tx 10 · daily 100 · weekly 700 · manual approval > 50.
Identical semantics, identical mental model — the skill drops into the AgentPay policy language with zero translation.

PART III

THE PROTOCOL STACK

▷ x402 · MPP · AP2 · A2A · MCP

x402

MACHINE-SPEED SETTLEMENT

Coinbase · 2025 · HTTP 402 revived · the payment handshake layer

01

REQUEST

Agent calls a gated endpoint

02

402

Server replies PAYMENT-REQUIRED

03

SIGN

Payload signed via KMS wallet

04

VERIFY

On-chain check — EIP-3009 / Permit2

05

DELIVER

200 + resource + receipt header

\$0.0001

per-tx settlement cost

< 2 sec

on-chain finality

154M

tx / yr across Base + Solana

\$600M

annualized volume

USD1 ANGLE — x402 settles in EIP-3009-capable stablecoins. USD1 on Ethereum, BNB Chain, and Solana can be the settlement asset behind every gated API, dataset, and inference call — regulated dollars at machine speed.

MPP

MACHINE PAYMENTS PROTOCOL

Stripe + Tempo · 2026 · IETF draft · the settlement + session layer

SESSIONS

Pre-authorize a budget once → stream thousands of signed micropayments off-chain → one on-chain settlement.

METHOD-AGNOSTIC

Stablecoins, fiat via Stripe Shared Payment Tokens, Lightning, cards — one HTTP interface for all rails.

NO FACILITATOR

Payment verification is a first-class protocol primitive — no third-party facilitator in the trust path.

x402-COMPATIBLE

Backward-compatible with x402's charge flow; MPP clients consume existing x402 services unchanged.

IN THE BOT + WLFI ANGLE — the skill speaks MPP today, as client and as server. MPP sessions are the natural spending-budget primitive for the WLFI App: authorize once, let an agent stream USD1 payments under policy — no checkout, no cart, no friction.

Day-one adopters: OpenAI · Anthropic · Google Gemini · Visa · Mastercard · Lightspark → 100+ listed services

AP2

AGENT PAYMENTS PROTOCOL

Google + 60 partners · 2025 · the trust / authorization layer — TradFi's rail

01

INTENT MANDATE

"Buy X under \$100."

User's initial intent — ceiling, timing, constraints. Signed upfront by the human.

02

CART MANDATE

Locks items + final price.

Merchant cryptographically signs the cart — a guarantee of fulfillment.

03

PAYMENT MANDATE

Authorizes settlement.

Shared with the network; signals agent-presence so issuers can price risk and route rails.

TRADFI ANGLE — Mandates form a non-repudiable evidence chain (intent → cart → payment) that regulators and card networks understand. AP2's partner set — Mastercard, Amex, PayPal, Coinbase — mirrors WLF's institutional orbit.

In the skill today: the AP2 rail drives Stripe · PayPal · Visa · Mastercard gateways under one policy engine.

A2A

AGENT-TO-AGENT PROTOCOL

Google · Linux Foundation · bot ↔ bot interactions & skills discovery

AGENT CARDS

JSON manifest at `/.well-known/agent.json` — the agent's business card: skills, endpoint, auth. This is how the payments bot gets discovered.

SKILLS DISCOVERY

Counterparty agents query capabilities before engaging — “can you settle USD1 over x402?” answered protocol-natively.

TASKS & PARTS

Stateful Task objects with lifecycles; multipart messages carry text, JSON, files across the agent mesh.

TRANSPORTS

HTTP + SSE + JSON-RPC with OAuth 2.0, PKCE, API keys out of the box. No reinvention.

UPCOMING — A2A support arrives in one of the bot's next releases: the skill publishes its Agent Card, exposes payment capabilities for discovery, and negotiates bot-to-bot settlements — paired with ERC-8004 identity (next section).

COMPOSABLE FINANCE

MONEY LEGOS FOR AGENTS — DISCOVER · COMPOSE · SETTLE

01

DISCOVER

Agents find financial skills through A2A Agent Cards — payments, FX, custody, lending, invoicing — queried protocol-natively, no directories, no integrations.

02

COMPOSE

Client + server duality makes skills stackable: every bot consumes services and provides its own. Financial workflows assemble like functions.

03

SETTLE

Composition ends in settlement — x402, MPP, or AP2, chosen per hop. USD1 as the common denominator keeps multi-agent flows in one compliant unit.

A COMPOSITION, END TO END —

A treasury agent discovers an FX-quotes bot and this payments bot via A2A · pulls conversion context · rebalances a USD1 position on WLFI Markets · pays suppliers over x402 · meters its own advisory service to other agents over MPP — every hop policy-gated and audit-logged.

Every agent is both a customer and a merchant. Server mode turns each skill into a revenue-bearing primitive; A2A discovery turns the mesh into a market.

MCP

MODEL CONTEXT PROTOCOL

Anthropic · the context layer — data & AI-system integration, NOT payments

△ **A DELIBERATE SEPARATION** — MCP moves context between models, data sources, and other AI systems. Value transfer lives in x402 / MPP / AP2. The bot keeps these planes apart by design.

FX-RATES ORACLE VIA MCP

Shipping with the bot's MCP support (upcoming release) — the flagship context integration:

- ▶ Live FX + crypto rates streamed into the agent's context window
- ▶ Policy engine normalizes limits across currencies (USD1 / EUR / BNB...)
- ▶ Conversion quotes priced before any payment intent is emitted
- ▶ Rates inform decisions — they never touch the settlement path

ONE PORT, MANY SOURCES

The same MCP port connects the bot to the wider world:

- ▶ Market data, chain indexers, compliance / sanctions feeds
- ▶ Other AI systems and agents exposed as MCP servers
- ▶ WLF internal systems: reserves dashboards, Markets state
- ▶ 97M SDK downloads/mo — every major client already speaks it

CONTEXT flows through MCP. VALUE flows through x402 / MPP / AP2. Never the same wire.

THE FULL STACK

FIVE PROTOCOLS · ONE SKILL · USD1 FLOWING THROUGH

L5

AP2

INTENT & TRUST

Mandates · verifiable credentials · accountability chain

IN THE BOT – client + server

L4

A2A

AGENT COLLABORATION

Agent cards · tasks · skills discovery

UPCOMING RELEASE

L3

MPP

SETTLEMENT & SESSIONS

Streaming micropayments · budgets · one API

IN THE BOT – client + server

L2

x402

PAYMENT HANDSHAKE

HTTP 402 · sub-2-second on-chain finality

IN THE BOT – client + server

L1

MCP

CONTEXT & TOOLS

Data access · tool invocation · FX-rates oracle

UPCOMING RELEASE

Context flows up. Money flows down. Policy gates everything.

PART IV

TRUST & SECURITY

▷ ERC-8004 · AGENTIC TRUST · SANDBOXING

ERC-8004

TRUSTLESS AGENTS

MetaMask · EF · Google · Coinbase — A2A's on-chain trust layer

IDENTITY REGISTRY

ERC-721 + URISStorage

Every agent is a portable, transferable NFT. tokenURI resolves to a registration file: endpoints, wallets, supported trust models. CAIP-10 identifiers make identity cross-chain.

REPUTATION REGISTRY

structured feedback

Authorized clients post bounded attestations — score, tags, evidence URI + hash — on-chain as composable signals. Aggregation and Sybil filtering happen off-chain.

VALIDATION REGISTRY

independent verification

Agents request validation of work; validator contracts respond via stake-secured re-execution, zkML verifiers, or TEE oracles. Results tracked on-chain.

DEPLOYED

Singleton 0x8004... registries live on Ethereum mainnet + 20+ networks; Identity & Reputation contracts audited

TRUST MODELS

Three composable models: reputation · crypto-economic validation · TEE attestation — pick per counterparty risk

IN THE BOT

Next releases: the skill registers its on-chain identity, accrues reputation per settled tx, and links TEE attestations to the Validation Registry

AGENTIC TRUST

PRINCIPLES — HOW AN AGENT EARNS THE RIGHT TO MOVE MONEY

DON'T TRUST — VERIFY

No pre-existing relationships assumed. Counterparties are checked against identity, reputation, and validation signals before value moves.

POLICY OVER DISCRETION

The LLM proposes; the deterministic policy engine disposes. Model output is never the last word on a transaction.

KEYS OUTSIDE THE MODEL

Private keys live in KMS / TEE — never in context, never in prompts. Prompt injection cannot exfiltrate what the model never holds.

LEAST PRIVILEGE

Scoped, revocable delegation: allowlists, budgets, time windows. An agent gets exactly the authority its task needs — no more.

VERIFIABLE AUDIT

Append-only logs plus on-chain receipts close the loop: every action feeds the reputation record the next counterparty reads.

HUMAN SOVEREIGNTY

Above materiality thresholds, a human signs. Autonomy is graduated, earned, and always revocable.

Trust = identity + policy + attestation + track record — all verifiable.

SANDBOXING THE AGENT

ISOLATION LADDER — MATCH THE BOUNDARY TO THE THREAT MODEL

L1 CONTAINERS

namespaces · seccomp · AppArmor

Shared host kernel — for trusted code. Ships today: Docker + compose.

L2 gVISOR

userspace kernel (Sentry)

Syscalls intercepted in user space; small host-syscall allowlist. Strong isolation, low overhead.

L3 microVM

Firecracker · Kata

Hardware virtualization, own guest kernel per agent; ~150 ms snapshot restore. The 2026 default for untrusted code.

L4 CVM / TEE

Nitro Enclaves · Intel TDX · AMD SEV-SNP

Encrypted memory + remote attestation — even the host can't peek. Signing and key custody live here.

ATTESTATION CLOSES THE LOOP — the TEE's remote-attestation quote is published to the ERC-8004 Validation Registry. Counterparties verify the agent runs the exact audited build before trusting a single payment — sandboxing becomes an on-chain trust signal.

UPCOMING RELEASES

WHAT SHIPS NEXT — THE NEXT THREE VERSIONS

AVAILABLE TODAY — x402 + MPP + AP2 as client and server · multi-backend router · policy engine · AWS KMS custody · audit trail · human-in-the-loop · SKILL.md for every major agent client.

RELEASE +1

A2A

- ▶ Agent Card published at /.well-known/agent.json
- ▶ Payment capabilities exposed for skills discovery
- ▶ Bot-to-bot settlement negotiation over A2A tasks

RELEASE +2

MCP

- ▶ FX-rates oracle served over MCP into agent context
- ▶ Market data, chain indexers, compliance feeds — one port
- ▶ Cross-currency policy normalization for limits & quotes

RELEASE +3

TRUSTLESS AGENTS

- ▶ ERC-8004 identity, reputation & validation registration
- ▶ Reputation accrual per settled transaction
- ▶ TEE attestation linked to the Validation Registry

All three tracks are in active development — capability commitments, not calendar commitments.

Open-source core (Apache-2.0) · co-developed adapters · WLFI keeps full custody and policy control throughout.

END OF TRANSMISSION

THE ASK

Make USD1 the reserve asset of the machine economy.

A co-built pilot: one skill, five protocols, three rails in both roles — policy-gated USD1 settlement.

RAILS

x402 · MPP · AP2

DISCOVERY

A2A · MCP

TRUST

ERC-8004 · TEE

The agents are here. The dollar is ready. Let's connect them.



SENTIENT-AGI

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**github.com/sentient-agi/agent-
payments-
bot**

